

HW 4

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```
source("506021334_stats102a_hw4.R")
```

Part 1: Dealing with Large Numbers

1a)

```
# e1 <- pqnumber(5, 5, 5, c())  
# e1 <- pqnumber(1, 5.5, 5, c())  
# e1 <- pqnumber(1, 5, 5.5, c())  
# e1 <- pqnumber(1, 5, 5, c(1, 2))  
# e1 <- pqnumber(1, 1, 1, c(1, 2, 3.1))
```

```
options(digits = 10)  
n1 <- pqnumber(1, 3, 4, c(1, 2, 3, 4, 5, 6, 7, 8))  
n2 <- pqnumber(1, 6, 0, c(3, 9, 5, 1, 4, 1, 3))  
n3 <- pqnumber(-1, 5, 1, c(2, 8, 2, 8, 1, 7, 2))  
is_pqnumber(n1)
```

```
## [1] TRUE
```

```
is_pqnumber(1)
```

```
## [1] FALSE
```

```
print(n1, DEC=TRUE)
```

```
## 87654.321  
## sign = 1  
## p = 3  
## q = 4  
## nums = 1 2 3 4 5 6 7 8
```

```
print(as_numeric(n2))
```

```
## [1] 3.141593
```

```
print(as_numeric(n3))
```

```
## [1] -27.18282
```

```
print(as_pqnumber(-12343.141, p = 3, q = 4), DEC=TRUE)
```

```
## [1] -12343141  
## -12343.141  
## sign = -1  
## p = 3  
## q = 4  
## nums = 1 4 1 3 4 3 2 1
```

1b) Addition and Subtraction

```
n1 <- pqnumber(1, 3, 4, c(1, 2, 3, 4, 5, 6, 7, 8))  
n2 <- pqnumber(1, 6, 0, c(3, 9, 5, 1, 4, 1, 9))  
  
# add(n1, n2)  
print(n1, DEC = TRUE)
```

```
## 87654.321  
## sign = 1  
## p = 3  
## q = 4  
## nums = 1 2 3 4 5 6 7 8
```

```
print(n2, DEC = TRUE)
```

```
## 9.141593  
## sign = 1  
## p = 6  
## q = 0  
## nums = 3 9 5 1 4 1 9
```

```
# align_pqnumber(n1, 6, 5)  
# align_pqnumber(n2, 6, 5)  
print(add(n1, n2), DEC = TRUE)
```

```
## 8768.346259  
## sign = 1  
## p = 6  
## q = 3  
## nums = 9 5 2 6 4 3 8 6 7 8
```

part c) Multiplication

```
print(multiply(n1, n2), DEC = TRUE)
```

```
## [1] 8.013001273e+15  
## 801300.1273  
## sign = 1  
## p = 10  
## q = 10  
## nums = 0 3 5 3 3 7 2 7 2 1 0 0 3 1 0 8 0 0 0 0 0
```